

T0 Theory: Calculation of Particle Masses and Physical Constants

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Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- Cosmological quantities (H_0 , Λ , critical density, age, Hubble length): removed from the table of constants and the category statistics; they are treated in dedicated documents (Docs. 064, 165, 279 for H_0 ; Doc. 308 for Λ). The previous deviations of 41–70 % rested on $H_0 \approx 40$ km/s/Mpc and contradicted the document's own footnote.
- Casimir-CMB ratio: ≈ 102.8 for the energy density instead of 308 (Casimir pressure); treated in Docs. 061, 166 and 279.
- Loschmidt constant: T0 value $2.687 \times 10^{25} \text{ m}^{-3}$ instead of 2.687×10^{22} (value per litre).
- Input values: ξ and three scale values for conversion (ℓ_P , E_0 , ν) instead of "three input values".
- Gravitation: $E_{\text{char}} = 28.4 \rightarrow 28.8$ (uniform with Doc. 012), factor 1 $3.521 \rightarrow 3.47222 \times 10^{-2}$, factor 2 $2.843 \rightarrow 2.88294 \times 10^{-5}$ (product and G_{SI} unchanged); $2(E_{\text{char}}\xi)^2$ now deviates by 2.3 %; consequential values r_0 , G_{char} , $260\times$.
- Summary "No arbitrary decimal adjustments": factor 2 is fitted to G ($2(E_{\text{char}}\xi)^2$ deviates by 2.3 %).
- G precision (note of 1 Oct. 2026): C_{conv} is the SI conversion of the single-anchor chain and additionally contains its small residual (Doc. 383); the agreement to 0.0125 % does not count as an independent precision result (Doc. 190, R141).

Abstract

The T0 theory presents a new approach to unifying particle physics and cosmology by deriving all fundamental masses and physical constants from the parameter $\xi = \frac{4}{3} \times 10^{-4}$ and three scale values for conversion: the Planck length $\ell_P = 1.616 \times 10^{-35} \text{ m}$, and the characteristic energy $E_0 = 7.398 \text{ MeV}$, where the energy can also be derived. This version demonstrates the remarkable precision of the T0 framework with over 99% accuracy for fundamental constants.

.1 Introduction

The T0 theory is based on the fundamental hypothesis of a geometric constant ξ that unifies all physical phenomena on macroscopic and microscopic scales. Unlike standard approaches based on empirical adjustments, T0 derives all parameters from exact mathematical relationships.

Fundamental Parameters

The entire T0 system is based on the parameter ξ and three scale values for the conversion to SI units: *(corrected on 30 Sept. 2026: previously "exclusively on three input values", although the list gives four values)*

$$\xi = \frac{4}{3} \times 10^{-4} \approx 1.33333333 \times 10^{-4} \quad (\text{geometric constant}) \quad (1)$$

$$\ell_P = 1.616 \times 10^{-35} \text{ m} \quad (\text{Planck length}) \quad (2)$$

$$E_0 = 7.398 \text{ MeV} \quad (\text{characteristic energy}) \quad (3)$$

$$v = 246.0 \text{ GeV} \quad (\text{Higgs vacuum expectation value}) \quad (4)$$

.2 T0 Fundamental Formula for the Gravitational Constant

Mathematical Derivation

The central insight of T0 theory is the relationship:

$$\xi = 2\sqrt{G \cdot m_{\text{char}}} \quad (5)$$

where $m_{\text{char}} = \xi/2$ is the characteristic mass. Solving for G yields:

$$G = \frac{\xi^2}{4m_{\text{char}}} = \frac{\xi^2}{4 \cdot (\xi/2)} = \frac{\xi}{2} \quad (6)$$

Dimensional Analysis

In natural units ($\hbar = c = 1$), the T0 fundamental formula initially gives:

$$[G_{\text{T0}}] = \frac{[\xi^2]}{[m]} = \frac{[1]}{[E]} = [E^{-1}] \quad (7)$$

Since the physical gravitational constant requires dimension $[E^{-2}]$, a conversion factor is necessary:

$$G_{\text{nat}} = G_{\text{T0}} \times 3.47222 \times 10^{-2} \quad [E^{-2}] \quad (8)$$

Origin of Factor 1 (3.47222×10^{-2})

(Corrected on 30 Sept. 2026: previously $E_{\text{char}} = 28.4$ and factor 1 = 3.521×10^{-2} , factor 2 = 2.843×10^{-5} – uniformly with Doc. 012, the derived value $E_{\text{char}} = 28.8$ now applies, i.e. factor 1 = $1/28.8 = 3.47222 \times 10^{-2}$. The product of both factors (1.00102×10^{-6}) stays the same, hence factor 2 = 2.88294×10^{-5} and G_{SI} unchanged. The assignment factor 2 = $2(E_{\text{char}}\xi)^2$ thus deviates by 2.3 % (previously 0.9 %); on this route factor 2 is an adjusted value. Consequential values: $r_0 = t_0 = 0.034722$, factor $\times \xi = 4.630 \times 10^{-6}$ ($260 \times \alpha_G$), $G_{\text{char}} = 6.0 \times 10^{-4}$.)

The factor 3.47222×10^{-2} originates from the characteristic T0 energy scale $E_{\text{char}} \approx 28.8$ in natural units. This factor corrects the dimension from $[E^{-1}]$ to $[E^{-2}]$ and represents the coupling of the T0 geometry to spacetime curvature as defined by the ξ -field structure.

Verification of the Characteristic T0 Factor

The factor 3.47222×10^{-2} is exactly $\frac{1}{28.8}$!

Key Results of Recalculation

1. Factor Identification:

- $3.47222 \times 10^{-2} = \frac{1}{28.8}$ (perfect agreement)
- This corresponds to a characteristic T0 energy scale of $E_{\text{char}} \approx \mathbf{28.8}$ in natural units

2. Dimensional Structure:

- $E_{\text{char}} = \mathbf{28.8}$ has dimension $[E]$
- Factor $= \frac{1}{28.8} \approx \mathbf{0.03472}$ has dimension $[E^{-1}] = [L]$
- This is a **characteristic length** in the T0 system

3. Dimensional Correction $[E^{-1}] \rightarrow [E^{-2}]$:

- Factor $\times = \mathbf{4.630 \times 10^{-6}}$ has dimension $[E^{-1}]$ (*Corrected on 29 Sept. 2026: previously "yields dimension $[E^{-2}]$ " – ξ is dimensionless, so the product keeps $[E^{-1}]$; the step to $[E^{-2}]$ happens in $G_{\text{nat}} = G_{T0} \times 3.47222 \times 10^{-2}$ with $[G_{T0}] = [E^{-1}]$.)*
- This is the coupling to spacetime curvature
- **260×** stronger than the pure gravitational coupling $\alpha_G = \xi^2 = 1.778 \times 10^{-8}$

4. Scale Hierarchy Confirmed:

$$E_0 \approx 7.398 \text{ MeV} \quad (\text{electromagnetic scale}) \quad (9)$$

$$E_{\text{char}} \approx 28.8 \quad (\text{T0 intermediate energy scale}) \quad (10)$$

$$E_{T0} = \frac{1}{\xi} = 7500 \quad (\text{fundamental T0 scale}) \quad (11)$$

5. **Physical Meaning:** The factor represents the ξ -**field structure coupling** that binds the T0 geometry to spacetime curvature - exactly as we described!

Formula for the characteristic T0 energy scale:

$$E_{\text{char}} = \frac{1}{3.47222 \times 10^{-2}} = 28.8 \quad (\text{natural units}) \quad (12)$$

The dimensional correction is achieved through the ξ -field structure:

$$\underbrace{3.47222 \times 10^{-2}}_{[E^{-1}]} \times \underbrace{\xi}_{[1]} = \underbrace{4.630 \times 10^{-6}}_{[E^{-1}]} \quad (13)$$

(*Corrected on 29 Sept. 2026: previously result dimension $[E^{-2}] = [E^{-1}] \cdot [1] = [E^{-1}]$.)* This coupling binds the T0 geometry to spacetime curvature.

Characteristic T0 Units: $r_0 = E_0 = m_0$

In characteristic T0 units of the natural unit system, the fundamental relationship holds:

$$r_0 = E_0 = m_0 \quad (\text{in characteristic units}) \quad (14)$$

Correct interpretation in natural units:

$$r_0 = 0.034722 \quad [E^{-1}] = [L] \quad (\text{characteristic length}) \quad (15)$$

$$E_0 = 28.8 \quad [E] \quad (\text{characteristic energy}) \quad (16)$$

$$m_0 = 28.8 \quad [E] = [M] \quad (\text{characteristic mass}) \quad (17)$$

$$t_0 = 0.034722 \quad [E^{-1}] = [T] \quad (\text{characteristic time}) \quad (18)$$

Fundamental conjugation:

$$r_0 \times E_0 = 0.034722 \times 28.8 = 1.000 \quad (\text{dimensionless}) \quad (19)$$

The characteristic scales are **conjugate quantities** of the T0 geometry. The T0 formula $r_0 = 2GE$ is used with the characteristic gravitational constant:

$$G_{\text{char}} = \frac{r_0}{2 \times E_0} = \frac{1}{2 \times E_{\text{char}}^2} \approx 6.0 \times 10^{-4} \quad (20)$$

(Corrected on 29 Sept. 2026: previously the second equality read $= \xi^2 / (2E_{\text{char}})$ – this gives 3.1×10^{-10} , whereas the left-hand side is $0.034722/57.6 = 6.0 \times 10^{-4}$; with $r_0 = 1/E_0$ one has $G_{\text{char}} = 1/(2E_{\text{char}}^2)$.)

SI Conversion

The transition to SI units is achieved through the conversion factor:

$$G_{\text{SI}} = G_{\text{nat}} \times 2.88294 \times 10^{-5} \quad \text{m}^3\text{kg}^{-1}\text{s}^{-2} \quad (21)$$

Origin of Factor 2 (2.88294×10^{-5})

The factor 2.88294×10^{-5} results from the fundamental T0 field coupling:

$$2.88294 \times 10^{-5} = 2 \times (E_{\text{char}} \times \xi)^2 \quad (22)$$

This formula has a clear physical meaning:

- **Factor 2:** Fundamental duality of T0 theory
- $E_{\text{char}} \times \xi$: Coupling of the characteristic energy scale to the ξ -geometry
- **Squaring:** Characteristic for field theories (analogous to E^2 terms)

Numerical verification:

$$2 \times (E_{\text{char}} \times \xi)^2 = 2 \times (28.8 \times 1.333 \times 10^{-4})^2 \quad (23)$$

$$= 2 \times (3.839 \times 10^{-3})^2 \quad (24)$$

$$= 2.948 \times 10^{-5} \quad (25)$$

Deviation from the used value: +2.3% (2.948 instead of 2.88294); the factor 2.88294×10^{-5} used in Step 4 thus follows from this formula only to about 2.3%.

(Corrected on 29 Sept. 2026: previously "< 1% (practically perfect agreement)" – with 2.948×10^{-5} instead of 2.88294×10^{-5} , G_{SI} would be 2.3% higher; the 0.0125% accuracy in Step 4 rests on the value 2.88294×10^{-5} , not on the formula.)

Step-by-Step Calculation

$$\text{Step 1: } m_{\text{char}} = \frac{\xi}{2} = \frac{1.333333 \times 10^{-4}}{2} = 6.666667 \times 10^{-5} \quad (26)$$

$$\text{Step 2: } G_{\text{T0}} = \frac{\xi^2}{4m_{\text{char}}} = \frac{\xi}{2} = 6.666667 \times 10^{-5} \quad [\text{E}^{-1}] \quad (27)$$

$$\text{Step 3: } G_{\text{nat}} = G_{\text{T0}} \times 3.47222 \times 10^{-2} = 2.314815 \times 10^{-6} \quad [\text{E}^{-2}] \quad (28)$$

$$\text{Step 4: } G_{\text{SI}} = G_{\text{nat}} \times 2.88294 \times 10^{-5} = 6.673469 \times 10^{-11} \text{m}^3\text{kg}^{-1}\text{s}^{-2} \quad (29)$$

(Corrected on 29 Sept. 2026: in Step 2 previously "[dimensionless]" – consistent with the dimensional analysis above, $[G_{T0}] = [E^{-1}]$.)

Experimental comparison:

$$G_{\text{exp}} = 6.674300 \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2} \quad (30)$$

$$\text{Relative deviation} = 0.0125\% \quad (31)$$

(Note of 1 Oct. 2026: The bridge factors 3.47222×10^{-2} and 2.88294×10^{-5} carry here the SI conversion of G that C_{conv} performs in Doc. 012. C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383, [K]); with the anchor m_e , G lies at -2.6% , with v at -0.46% (Doc. 383). The agreement to 0.0125% (with C_{conv} : 0.003%) arises only with this residual and does not count as an independent precision result (Doc. 190, R141).)

.3 Particle Mass Calculations

T0 Yukawa Method

All fermion masses are determined by the universal T0 Yukawa formula:

$$m = r \times \xi^p \times v \quad (32)$$

where r and p are exact rational numbers following from the T0 geometry.

Detailed Mass Calculations

Table 1: T0 Yukawa mass calculations for all Standard Model fermions

Particle	r	p	ξ^p	T0 Mass [MeV]	Exp. [MeV]	Error [%]
Electron	$\frac{4}{3}$	$\frac{3}{2}$	1.540e-06	0.505	0.511	1.18
Muon	$\frac{16}{5}$	1	1.333e-04	105.0	105.7	0.66
Tau	$\frac{25}{9}$	$\frac{2}{3}$	2.610e-03	1783.4	1776.9	0.37
Up	6	$\frac{3}{2}$	1.540e-06	2.272	2.270	0.11
Down	$\frac{25}{2}$	$\frac{3}{2}$	1.540e-06	4.734	4.720	0.30
Strange	$\frac{26}{9}$	1	1.333e-04	94.8	93.4	1.45
Charm	2	$\frac{2}{3}$	2.610e-03	1284.1	1270.0	1.11
Bottom	$\frac{3}{2}$	$\frac{1}{2}$	1.155e-02	4260.8	4180.0	1.93
Top	$\frac{1}{28}$	$\frac{-1}{3}$	1.957e+01	171974.5	172760.0	0.45

Example Calculation: Electron

The electron mass serves as a paradigmatic example of the T0 Yukawa method:

$$r_e = \frac{4}{3}, \quad p_e = \frac{3}{2} \quad (33)$$

$$m_e = \frac{4}{3} \times \left(\frac{4}{3} \times 10^{-4} \right)^{3/2} \times 246 \text{ GeV} \quad (34)$$

$$= \frac{4}{3} \times 1.539601 \times 10^{-6} \times 246 \text{ GeV} \quad (35)$$

$$= 0.505 \text{ MeV} \quad (36)$$

Experimental value: $m_{e,\text{exp}} = 0.511 \text{ MeV}$

Relative deviation: 1.176%

.4 Magnetic Moments and g-2 Anomalies

Quantitative results and comparison tables for leptonic anomalous magnetic moments are centralized in the dedicated document 018_T0_Anomalous-g2-10_En.pdf. This overview of complete calculations only notes that such tests exist and refers the reader to that document for explicit values and detailed analyses.

.5 Complete List of Physical Constants

The T0 theory calculates over 40 fundamental physical constants in a hierarchical 8-level structure. This section documents all calculated values with their units and deviations from experimental reference values.

Categorized Constants Overview

Category	Count	Avg. Error [%]	Min [%]	Max [%]	Precision
Fundamental	1	0.0005	0.0005	0.0005	Excellent
Gravitation	1	0.0125	0.0125	0.0125	Excellent
Planck	6	0.0083	0.0062	0.0125	Excellent
Electromagnetic	4	0.0001	0.0000	0.0002	Excellent
Atomic Physics	7	0.0005	0.0000	0.0009	Excellent
Metrology	5	0.0002	0.0000	0.0005	Excellent
Thermodynamics	3	0.0008	0.0000	0.0023	Excellent
Cosmology	–	dedicated documents*			

Table 2: Category-based error statistics of T0 constant calculations

Detailed Constants List

Table 3: Complete list of all calculated physical constants

Constant	Sym- bol	T0 Value	Reference Value	Error [%]	Unit
Fine-structure constant	α	7.297e-03	7.297e-03	0.0005	dimensionless
Gravitational constant	G	6.673e-11	6.674e-11	0.0125	$\text{m}^3\text{kg}^{-1}\text{s}^{-2}$
Planck mass	m_P	2.177e-08	2.176e-08	0.0062	kg
Planck time	t_P	5.391e-44	5.391e-44	0.0062	s
Planck temperature	T_P	1.417e+32	1.417e+32	0.0062	K
Speed of light	c	2.998e+08	2.998e+08	0.0000	m/s
Reduced Planck constant	$\hbar$	1.055e-34	1.055e-34	0.0000	J s
Planck energy	E_P	1.956e+09	1.956e+09	0.0062	J
Planck force	F_P	1.210e+44	1.210e+44	0.0125	N
Planck power	P_P	3.629e+52	3.628e+52	0.0125	W
Magnetic constant	μ_0	1.257e-06	1.257e-06	0.0000	H/m
Electric constant	ϵ_0	8.854e-12	8.854e-12	0.0000	F/m
Elementary charge	e	1.602e-19	1.602e-19	0.0002	C
Impedance of free space	Z_0	3.767e+02	3.767e+02	0.0000	Ω
Coulomb constant	k_e	8.988e+09	8.988e+09	0.0000	Nm^2/C^2
Stefan-Boltzmann constant	σ_{SB}	5.670e-08	5.670e-08	0.0000	$\text{W}/\text{m}^2\text{K}^4$
Wien displacement constant	b	2.898e-03	2.898e-03	0.0023	m K
Planck constant	h	6.626e-34	6.626e-34	0.0000	J s
Bohr radius	a_0	5.292e-11	5.292e-11	0.0005	m
Rydberg constant	R_∞	1.097e+07	1.097e+07	0.0009	m^{-1}
Bohr magneton	μ_B	9.274e-24	9.274e-24	0.0002	J/T
Nuclear magneton	μ_N	5.051e-27	5.051e-27	0.0002	J/T
Hartree energy	E_h	4.360e-18	4.360e-18	0.0009	J
Compton wavelength	λ_C	2.426e-12	2.426e-12	0.0000	m
Classical electron radius	r_e	2.818e-15	2.818e-15	0.0005	m
Faraday constant	F	9.649e+04	9.649e+04	0.0002	C/mol
von Klitzing constant	R_K	2.581e+04	2.581e+04	0.0005	Ω
Josephson constant	K_J	4.836e+14	4.836e+14	0.0002	Hz/V
Magnetic flux quantum	Φ_0	2.068e-15	2.068e-15	0.0002	Wb
Gas constant	R	8.314e+00	8.314e+00	0.0000	J K/mol
Loschmidt constant	n_0	2.687e+25	2.687e+25	0.0000	m^{-3}

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Constant	Sym- bol	T0 Value	Reference Value	Error [%]	Unit
Cosmological quantities	$H_0, \Lambda, \rho_{\text{crit}}$	see Docs. 064, 165, 279 (H_0) and 308 (Λ)			
* Boltzmann constant	k_B	1.381e-23	1.381e-23	0.0000	J/K
Avogadro constant	N_A	6.022e+23	6.022e+23	0.0000	mol ⁻¹

(Corrected on 30 Sept. 2026: Loschmidt constant previously T0 value 2.687×10^{22} and error 99.9% – that is the value per litre; per m^3 it is 2.687×10^{25} . For electron, up and down the masses are now given with more digits; the error percentages refer to the anchors 0.511, 2.270 and 4.720 MeV from `calc_De.py`.)

(Corrected on 29 Sept. 2026: previously errors of 0.0158% for t_P and 0.0220% for F_P, P_P , and in the category table Planck mean 0.0131%, max 0.0220% – with $G_{T0}/G_{\text{exp}} - 1 = -0.0125\%$ one obtains 0.0062% for $t_P \propto \sqrt{G}$ and 0.0125% for $F_P, P_P \propto 1/G$; T0 values of t_P and F_P rounded accordingly.)

*Cosmological quantities: H_0, Λ and the critical density are treated in dedicated documents (Docs. 064, 165, 279 for H_0 ; Doc. 308 for Λ); the cosmic sector is deliberately excluded from the derivation of constants (Doc. 190, P39). In T0 the universe is static – age, critical density and Hubble length are Friedmann concepts without direct T0 counterpart.

(Corrected on 30 Sept. 2026: previously the table listed $\Lambda = 1.610 \times 10^{-52} \text{ m}^{-2}$ (45.7%), $\rho_{\text{crit}} = 3.001 \times 10^{-27} \text{ kg/m}^3$ (65.0%), the age of the universe (∞) and the Hubble length $2.314 \times 10^{26} \text{ m}$ (69.7%), and the category statistics “Cosmology: 4 entries, mean 55.2%, min 41.0%, max 69.7%”; these values rest on $H_0 \approx 40 \text{ km/s/Mpc}$. The footnote, by contrast, gave $E_H = E_0 \xi^{41/4} = 1.41 \times 10^{-33} \text{ eV}$ and $H_0 \approx 66.2 \text{ km/s/Mpc}$.)

.6 Mathematical Elegance and Theoretical Significance

Exact Rational Ratios

A remarkable property of T0 theory is the exclusive use of **exact mathematical constants**:

- **Basic constant:** $\xi = \frac{4}{3} \times 10^{-4}$ (exact fraction)
- **Particle r parameters:** $\frac{4}{3}, \frac{16}{5}, \frac{25}{9}, \frac{25}{2}, \frac{26}{9}, \frac{3}{2}, \frac{1}{28}$
- **Particle p parameters:** $\frac{3}{2}, 1, \frac{2}{3}, \frac{1}{2}, -\frac{1}{3}$
- **Gravitational factors:** $\frac{\xi}{2}, 3.47222 \times 10^{-2}, 2.88294 \times 10^{-5}$

The relationships follow from the geometric structure; an exception is factor 2 (2.88294×10^{-5}), which follows from $2(E_{\text{char}}\xi)^2 = 2.949 \times 10^{-5}$ only to 2.3% and is fitted to G (section on the origin of factor 2). (Corrected on 30 Sept. 2026: previously “No arbitrary decimal adjustments! All relationships follow from the fundamental geometric structure” – this contradicted the document’s own statement on factor 2.)

Dimension-Based Hierarchy

The T0 constant calculation follows a natural 8-level hierarchy:

1. **Level 1:** Primary ξ derivatives (α, m_{char})
2. **Level 2:** Gravitational constant (G, G_{nat})
3. **Level 3:** Planck system (m_P, t_P, T_P , etc.)

4. **Level 4:** Electromagnetic constants (e, ϵ_0, μ_0)
5. **Level 5:** Thermodynamic constants (σ_{SB} , Wien constant)
6. **Level 6:** Atomic and quantum constants (a_0, R_∞, μ_B)
7. **Level 7:** Metrological constants (R_K, K_J , Faraday constant)
8. **Level 8:** Cosmological constants (H_0, Λ , critical density) – in dedicated documents (064, 165, 279, 308)

Fundamental Significance of Conversion Factors

The conversion factors in the T0 gravitational calculation have profound theoretical significance:

$$\text{Factor 1: } 3.47222 \times 10^{-2} \quad [E^{-1} \rightarrow E^{-2}] \quad (37)$$

$$\text{Factor 2: } 2.88294 \times 10^{-5} \quad [E^{-2} \rightarrow m^3 kg^{-1} s^{-2}] \quad (38)$$

Interpretation: These factors do not arise from arbitrary adjustment, but represent the fundamental geometric structure of the ξ -field and its coupling to spacetime curvature.

Experimental Testability

The T0 theory makes specific, testable predictions:

1. **Casimir-CMB ratio:** see Docs. 061, 166 and 279; at $d \approx 100 \mu m$, $|\rho_{\text{Casimir}}|/\rho_{\text{CMB}} = \pi^2/(720 \xi) \approx 102.8$ (corrected on 30 Sept. 2026: previously ≈ 308 , the ratio for the Casimir pressure; already corrected in Docs. 061 and 063)
2. **Precise g-2 measurements:** T0 corrections for electron and tau
3. **Fifth force:** Modifications of Newtonian gravity on ξ -characteristic scales
4. **Cosmological parameters:** Alternative to Λ -CDM with ξ -based predictions (Docs. 064, 165, 279, 308)

.7 Methodological Aspects and Implementation

Numerical Precision

The T0 calculations consistently use:

- **Exact fraction calculations:** Python fractions.Fraction for r and p parameters
- **CODATA 2018 constants:** All reference values from official sources
- **Dimensional validation:** Automatic checking of all units
- **Error filtering:** Intelligent handling of outliers and T0-specific constants

Category-Based Analysis

The 40+ calculated constants are divided into physically meaningful categories:

Fundamental	α, m_{char} (directly from ξ)
Gravitation	G, G_{nat} , conversion factors
Planck	$m_P, t_P, T_P, E_P, F_P, P_P$
Electromagnetic	$e, \epsilon_0, \mu_0, Z_0, k_e$
Atomic Physics	$a_0, R_\infty, \mu_B, \mu_N, E_h, \lambda_C, r_e$
Metrology	$R_K, K_J, \Phi_0, F, R_{\text{Gas}}$
Thermodynamics	σ_{SB} , Wien constant, h
Cosmology	$H_0, \Lambda, t_{\text{Universe}}, \rho_{\text{crit}}$

.8 Comparison with Standard Approaches

Advantages of T0 Theory

1. **Parameter reduction:** the parameter ξ and three conversion scales instead of > 20 parameters in the Standard Model
2. **Mathematical elegance:** Exact fractions instead of empirical adjustments
3. **Unification:** Particle physics + cosmology + quantum gravity
4. **Predictive power:** New phenomena (Casimir-CMB, modified g-2)
5. **Experimental testability:** Specific, falsifiable predictions

Theoretical Challenges

1. **Conversion factors:** Theoretical derivation of numerical factors
2. **Quantization:** Integration into a complete quantum field theory
3. **Renormalization:** Treatment of divergences and scale invariances
4. **Symmetries:** Connection to known gauge symmetries
5. **Dark matter/energy:** Explicit T0 treatment of cosmological puzzles

.9 Technical Implementation Details

Python Code Structure

The T0 calculation program T0_calc_En.py is implemented as an object-oriented Python class:

```
class T0UnifiedCalculator:
    def __init__(self):
        self.xi = Fraction(4, 3) * 1e-4 # Exact fraction
        self.v = 246.0 # Higgs-VEV [GeV]
        self.l_P = 1.616e-35 # Planck length [m]
        self.E0 = 7.398 # Characteristic energy [MeV]

    def calculate_yukawa_mass_exact(self, particle_name):
        # Exact fraction calculations for r and p
        # T0 formula: m = r \times \xi^p \times v

    def calculate_level_2(self):
        # Gravitational constant with factors
        # G = \xi^2/(4m) \times 3.47222e-2 \times 2.88294e-5
```

Quality Assurance

- **Dimensional validation:** Automatic checking of all physical units
- **Reference value verification:** Comparison with CODATA 2018 and Planck 2018
- **Numerical stability:** Use of `fractions.Fraction` for exact arithmetic
- **Error handling:** Intelligent handling of T0-specific vs. experimental constants

.10 Appendix: Complete Data References

Experimental Reference Values

All experimental values used in this report come from the following authorized sources:

- **CODATA 2018:** Committee on Data for Science and Technology, "2018 CODATA Recommended Values"
- **PDG 2020:** Particle Data Group, "Review of Particle Physics", Prog. Theor. Exp. Phys. 2020
- **Planck 2018:** Planck Collaboration, "Planck 2018 results VI. Cosmological parameters"
- **NIST:** National Institute of Standards and Technology, Physics Laboratory

Software and Calculation Details

- **Python version:** 3.8+
- **Dependencies:** `math`, `fractions`, `datetime`, `json`
- **Precision:** Floating point: IEEE 754 double precision
- **Fraction calculations:** `Python fractions.Fraction` for exact arithmetic
- **Code repository:**